

Week 3 – Linear or Non-linear (Answers)

Geophysics IIIA: Potential Fields & Geothermics

Student Name: _____ Group: _____

Answer key (concise)

- A1. *Bilinear* in $(\Delta\rho, H)$; *linear* if one is known; or reparameterize $m = \Delta\rho H$.
- A2. *Non-linear* in (R, z) ; *linear* in $\Delta\rho$ (or M). Linearize via Jacobian about p_0 .
- A3. *Linear* in $[a, b, c]$ for fixed n and known X ; *non-linear* in n (block LS + GN).
- A4. *Linear* in k_0 if (A, B, c) known; *non-linear* jointly. Often fix (A, B) .
- A5. *Linear* relation between h and Δt for known densities; becomes *bilinear* if densities are unknown.
- A6. *Linear* (didactic column model) in the stated terms.
- A7. *Linear* in amplitude C (geometry fixed); *non-linear* in (x_1, x_2, z) .
- A8. *Linear* in heat production set $\{H_k\}$ for fixed $\{k_k\}$ once interface/BC constraints are applied; *non-linear* if k depends on T .
- A9. *Linear* in porosity ϕ for known (ρ_m, ρ_f) .
- A10. Non-linear. Take logs: $\ln \rho = \ln a + b \ln v$ (linear in $[\ln a, b]$).
- A11. *Linear* in $\Delta\rho_i$ (geometry fixed).